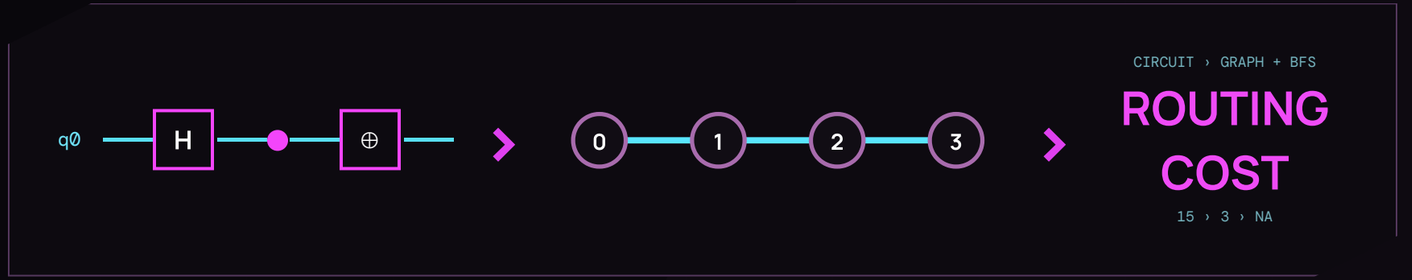
PQC-GRAPH-C

A C-Native Quantum Circuit Workbench

From circuit simulation to hardware-aware reasoning.

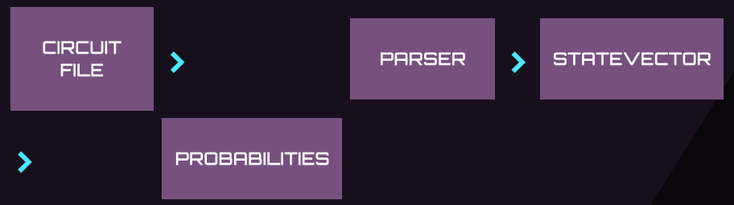
CIRCUIT → GRAPH + BFS → ROUTING COST



THE SIMULATOR ESTABLISHED CIRCUIT-LEVEL BEHAVIOR

CIRCUIT FILE · ORDERED GATES

```
4
H 0
RY 1 1.5708
CX 0 3
RZ 2 0.7854
```



THE NEXT ENGINEERING QUESTION

A mathematically valid circuit may request a two-qubit interaction that the selected hardware topology does not directly support.

LOGICAL REQUEST: CX 0 3



SIX GATES TRANSFORM ONE STATEVECTOR

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$P(0) = |\alpha|^2$, $P(1) = |\beta|^2$

$|\alpha|^2 + |\beta|^2 = 1$

X
Bit flip

`[0 1; 1 0]`

H
Superposition

`1/√2 [1 1; 1 -1]`

RX(θ)
X-axis rotation

`angle in radians`

RY(θ)
Y-axis rotation

`also used for encoding`

RZ(θ)
Relative-phase

`phase rotation`

CX
Controlled flip

`involves two qubits`



Rotation angles are in radians. **CX** is the connection between quantum-circuit logic and my graph-analysis problem.

LOGICAL INSTRUCTIONS MEET PHYSICAL CONNECTIVITY

```
Circuit requests:
CX 0 3
Selected hardware:
0 — 1 — 2 — 3
```



Direct edge 0-3? NO · Shortest path: 0 → 1 → 2 → 3

LOGICAL	PHYSICAL	ROUTING
requested gate	available edges	added movement

Qiskit

Quantum SDK, hardware-aware transpilation and routing

qsim

High-performance statevector simulation

QuEST

High-performance simulation, with parallel and accelerated execution options

PQC-GRAPH-C

Inspectable C implementation linking circuit instructions, adjacency lists, BFS paths and explicit cost estimates

My focus: transparent implementation and reproducible hardware-connectivity reasoning.

IBM Quantum qsim QuEST

ONE VALIDATED CIRCUIT. TWO INDEPENDENT ANALYSES.

CIRCUIT FILE

TOPOLOGY FILE

COMPLETE INPUT VALIDATION › VALIDATED INPUTS

VALIDATED CIRCUIT

SEPARATE PHYSICAL GRAPH

SIMULATION

Initialize statevector

Apply gates

Check norm

Probabilities

HARDWARE ANALYSIS

Extract CX operations

Check adjacency

Run BFS Classify CX

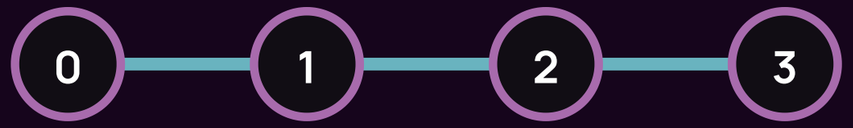
Estimate cost

CLI / WORKBENCH / ACTUAL PROGRAM OUTPUT

Critical invariant: a CX instruction never creates a hardware edge.

GRAPH + BFS EXPOSES THE SHORTEST HARDWARE ROUTE

REQUEST CX 0 3 › BFS from 0 to 3




```
adj[0] = {1}        adj[1] = {0, 2}
adj[2] = {1, 3}     adj[3] = {2}
```

Vertex physical qubit · **Edge** direct connection · **Distance** 3 edges

step	pop	queue[]	visited[]	parent[]
0	-	0	0	parent[0] = -1
1	0	1	0 1	parent[1] = 0
2	1	2	0 1 2	parent[2] = 1
3	2	3	0 1 2 3	parent[3] = 2 · target reached

RECONSTRUCT $3 \leftarrow 2 \leftarrow 1 \leftarrow 0$

SHORTEST PATH $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

distance = 3

visited[] prevents repeated visits · **queue[]** explores in breadth-first order · **parent[]** reconstructs the path. BFS guarantees the shortest path in unweighted graphs.

$O(V + E)$

ONE CIRCUIT. THREE TOPOLOGIES. THREE OUTCOMES.

RESTORE-3CX-V1

SWAPs = 2(d - 1)

Extra CX = 6(d - 1)

Total CX = 1 + 6(d - 1)



Single gate CX(0,3) (d = 3):
4 estimated SWAPs · 12 extra CX · 13 total CX equivalents

LINE: 0–1–2–3



15

dist: 1, 3, 1

4 SWAPs | 12 extra CX

SHORTCUT: ADD 0–3



3

dist: 1, 1, 1

0 SWAPs | 0 extra CX

DISCONNECTED: 0–1 2–3



NA

dist: 1, NA, 1

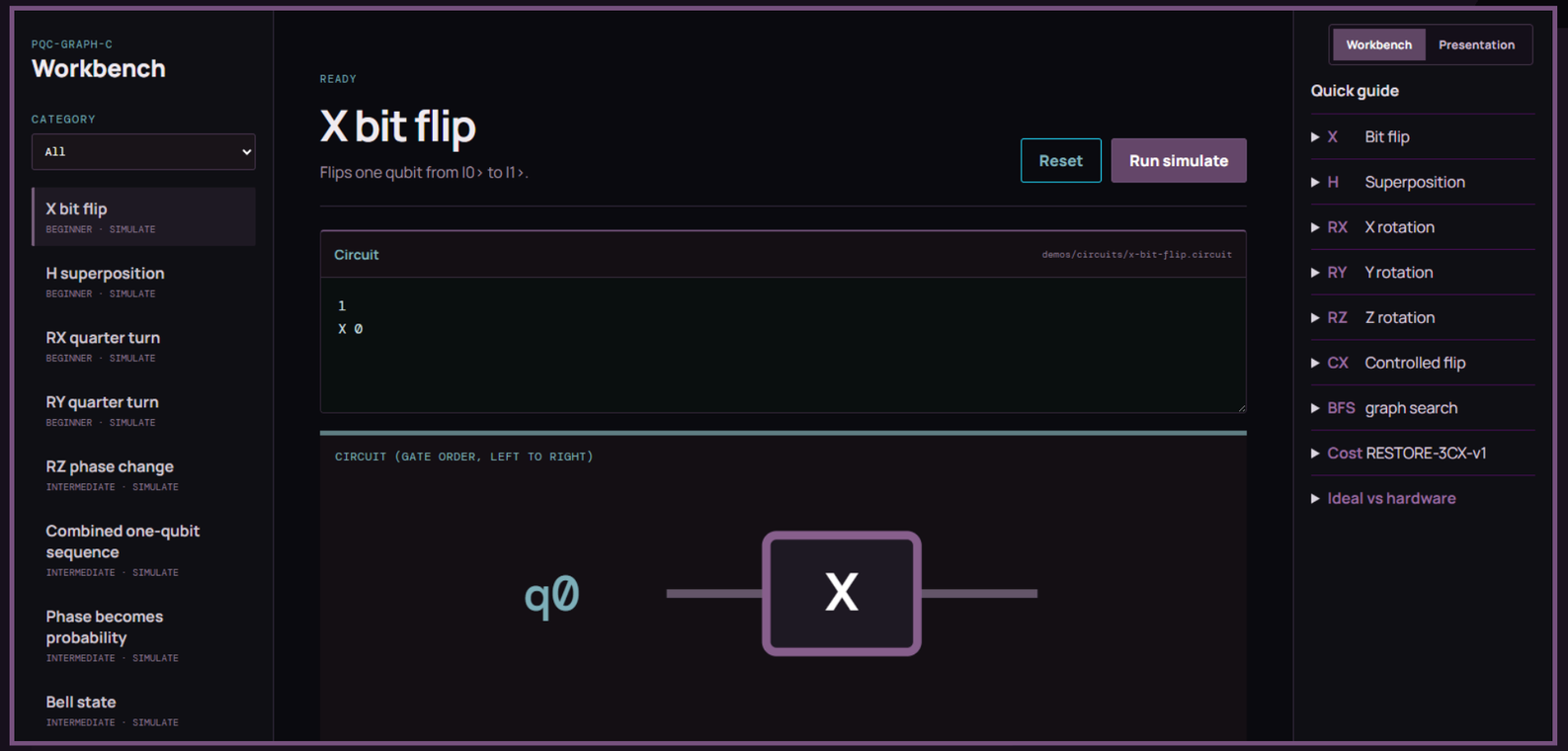
complete cost unavailable

15 › 3 › NA

Connectivity changes estimated cost or makes an interaction unreachable

Modeled gate counts under RESTORE-3CX-v1, not hardware runtime or optimal-routing measurements. For the 3-CX circuit: 1 + 13 + 1 = 15.

THE WORKBENCH EXPOSES THE C ENGINE



Actual local Workbench connected to the compiled C production executable

01 Load three-CX circuit

02 Select line topology

03 Run compiled C application

04 Inspect BFS path and estimated cost

05 Switch to shortcut topology and rerun

06 Try disconnected topology (NA)

07 Inspect graph.c & analysis.c in C source

```
classification = INDIRECT
path = 0 -> 1 -> 2 -> 3
distance = 3
estimated_swaps = 4
extra_cx_equivalents = 12
total_cx_equivalents = 13
```

A FOUNDATION FOR HARDWARE-AWARE REASONING

IMPLEMENTED

C GATE SIMULATOR

Validated quantum gate simulation and statevector probabilities.

IMPLEMENTED

GRAPH + BFS ANALYZER

Adjacency-list graph, deterministic BFS, direct/indirect/unreachable classification.

IMPLEMENTED

COST MODEL + WORKBENCH

RESTORE-3CX-v1 estimation, reproducible topology comparison and interactive workbench.

FUTURE ENGINEERING

- Actual circuit transformation with SWAP insertion and updated qubit mapping
- More advanced placement and routing strategies
- Reproducible performance benchmarks
- Investigation of optional GPU-accelerated classical simulation

Don't just simulate the circuit. Understand its relationship with the machine it is meant to run on.

SOURCES AND REPRODUCIBLE PROJECT EVIDENCE

IBM Quantum
Transpilation stages, layout and routing
`qiskit.qotlabs.org/docs/guides/transpile`

Google Quantum AI
qsim statevector simulator
`github.com/quantumlib/qsim`

QuEST
Quantum Exact Simulation Toolkit
`quest.qtechtheory.org`

Project evidence
Approved project specifications (DOC-04 SRS, DOC-05 SDD, DOC-07)
`docs/`

Production source
Production C codebase and test suite
`graph.c, analysis.c, sim.c, input.c, qml.c`

Execution records
Actual topology-analysis execution records (15, 3, NA)
`build/evidence/ & docs/demo-evidence/`